NEW VISION
THINK FORWARD
A SoftServe Company

CASE STUDY

AI-Agent Driven Contract Testing
for a Big 4 US Partnership Tax Platform



Client Overview

A Big 4 global professional services firm modernizing its US Partnership Tax platform, a microservices-based ecosystem of interdependent modules spanning client onboarding, data setup and allocations, federal and state tax calculation, and reports and forms generation. Accurate data exchange across these services is critical to compliance and timely filing.

Problem Statement

Defects escaped to production due to the lack of end-to-end, user-journey based testing, as well as ineffective unit and integration tests early in the release cycles. Key challenges included:

Production Defect Leakage

Integration defects surfaced late in UAT and production, where they are most costly to diagnose and fix.

No E2E User Journey Testing

Modules were tested in isolation; journeys from onboarding to tax calculation and forms were never continuously proven.

Ineffective Unit & Integration Tests

Mocks and stubs drifted from live API behavior, so green runs early in the cycle gave false confidence.

Undetected Contract Drift

Field, type, endpoint and payload changes reached downstream teams late, with costly manual upkeep of test expectations.

Our Solution

NewVision deployed its **AI-Agent Powered Contract Testing Framework**, a contract assurance layer on Pact where AI agents detect API drift and maintain tests under human approval, and CI verifies every contract.

Key Solution Components



Contract-as-Code Foundation

Pact V4 consumer-driven contracts for REST and GraphQL APIs, executed with Jest and TypeScript, with Zod schema validation.



Spec Retrieval Agent

Pulls the latest provider OpenAPI and Swagger specs from CI on every change, keeping contracts anchored to the source of truth.



Drift Detection Agent

Compares specs against Pact matchers, tests and Zod schemas, and produces a discrepancy report for each consumer and provider pair.



Change Classification Agent

Classifies additions, removals, renames, type shifts and endpoint changes as breaking or non-breaking to focus review effort.



Test Generation Agent

Generates missing tests, updates matchers and regenerates schemas; owners review every diff before anything is merged.



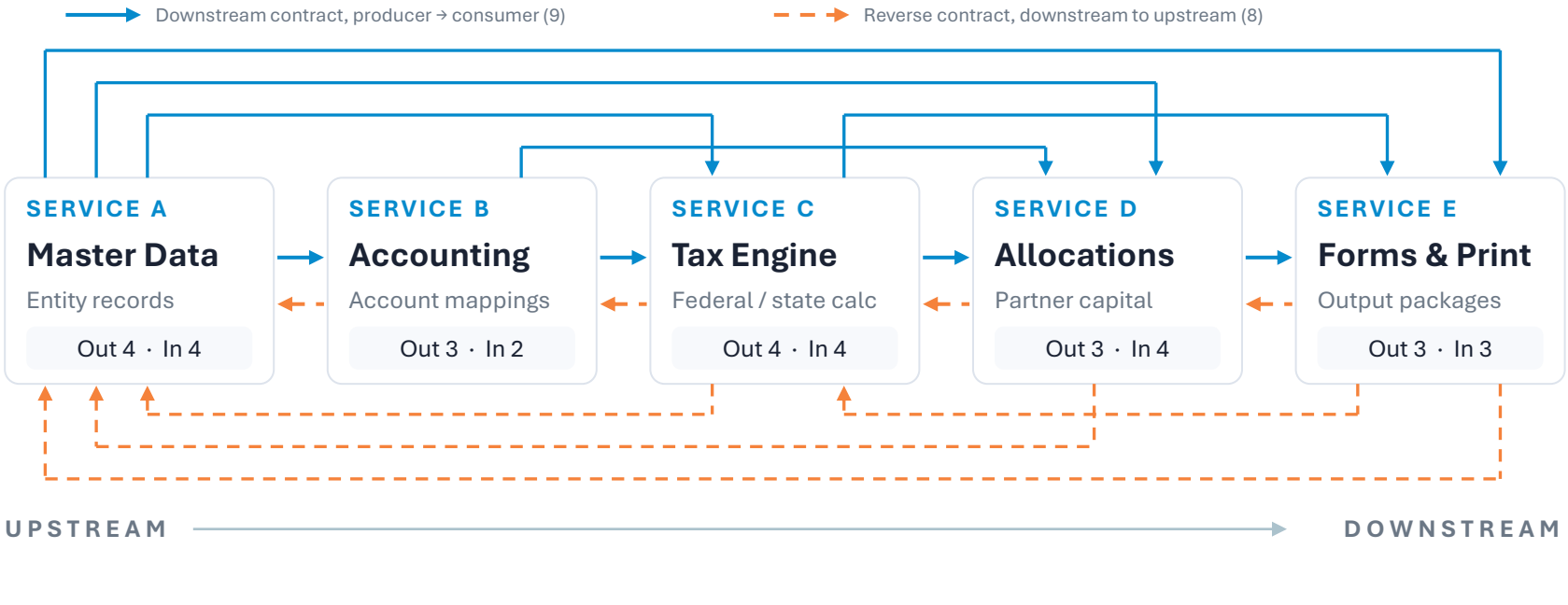
CI Verification & Governance

Pact provider checks run against live APIs in every pipeline, publishing pass/fail, change severity and an audit-ready evidence trail.

Control principle: AI proposes • Owners decide • Pact verifies • CI enforces

Contract Testing Architecture

Pact validates 17 producer to consumer contracts across five microservice domains; AI agents keep them current.



Contract matrix

Producer ↓ Consumer → (CT-nn)

	A	B	C	D	E
A Master Data	·	CT-01	CT-02	CT-03	CT-04
B Accounting	CT-05	·	CT-06	CT-07	n/a
C Tax Engine	CT-08	CT-09	·	CT-10	CT-11
D Allocations	CT-12	n/a	CT-13	·	CT-14
E Forms & Print	CT-15	n/a	CT-16	CT-17	·

Blue downstream **Orange** reverse

PACT TEST SUITE
CT-01 to CT-17 · Pact V4 · Jest / TS · Zod

Consumer tests
Expectations per contract

Pact Broker
Versioned contracts

Live verification
Producer APIs replayed

CI quality gate
Blocks breaking changes

updates contracts

producer specs from CI

AI AGENT LAYER
AI proposes, owners decide, Pact verifies, CI enforces

AGENT 4
Test Generation
Updates tests, matchers, schemas

HUMAN GATE
Owner Review
Approve, reject or investigate

AGENT 3
Classification
Breaking or non-breaking change

AGENT 2
Drift Detection
Specs vs matchers, tests, Zod

AGENT 1
Spec Retrieval
Pulls latest OpenAPI / Swagger from CI

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Business Outcomes & Benefits Delivered

Impact Area	Before Implementation	After Implementation	Business Benefit
Contract Upkeep Effort	7.5 hrs per API change, fully manual	1.75 hrs per change, AI-assisted	77% effort reduction
Effort per 100 API Changes	750 hrs	175 hrs	575 hrs released for higher-value testing
Drift Detection	Late, in integration or production support	On every CI run, before deployment	Breaking changes caught pre-merge
Integration Defect Leakage	Defects escaping to UAT and production	Incompatible changes blocked at CI gate	Fewer production escapes, lower fix cost
Release Confidence	Unit pass rates without compatibility proof	CI evidence: pass/fail, severity, audit trail	Evidence-led release decisions

Effort figures reflect NewVision's AI-assisted planning model per 100 contract-change events.

Strategic Value

01 Shifted integration defect detection left, catching breaking API changes before deployment, not in production.

02 Turned contract upkeep into exception-led assurance: AI agents draft updates; owners retain approval control.

03 Established shared, traceable contract ownership between consumer and provider teams across tax services.

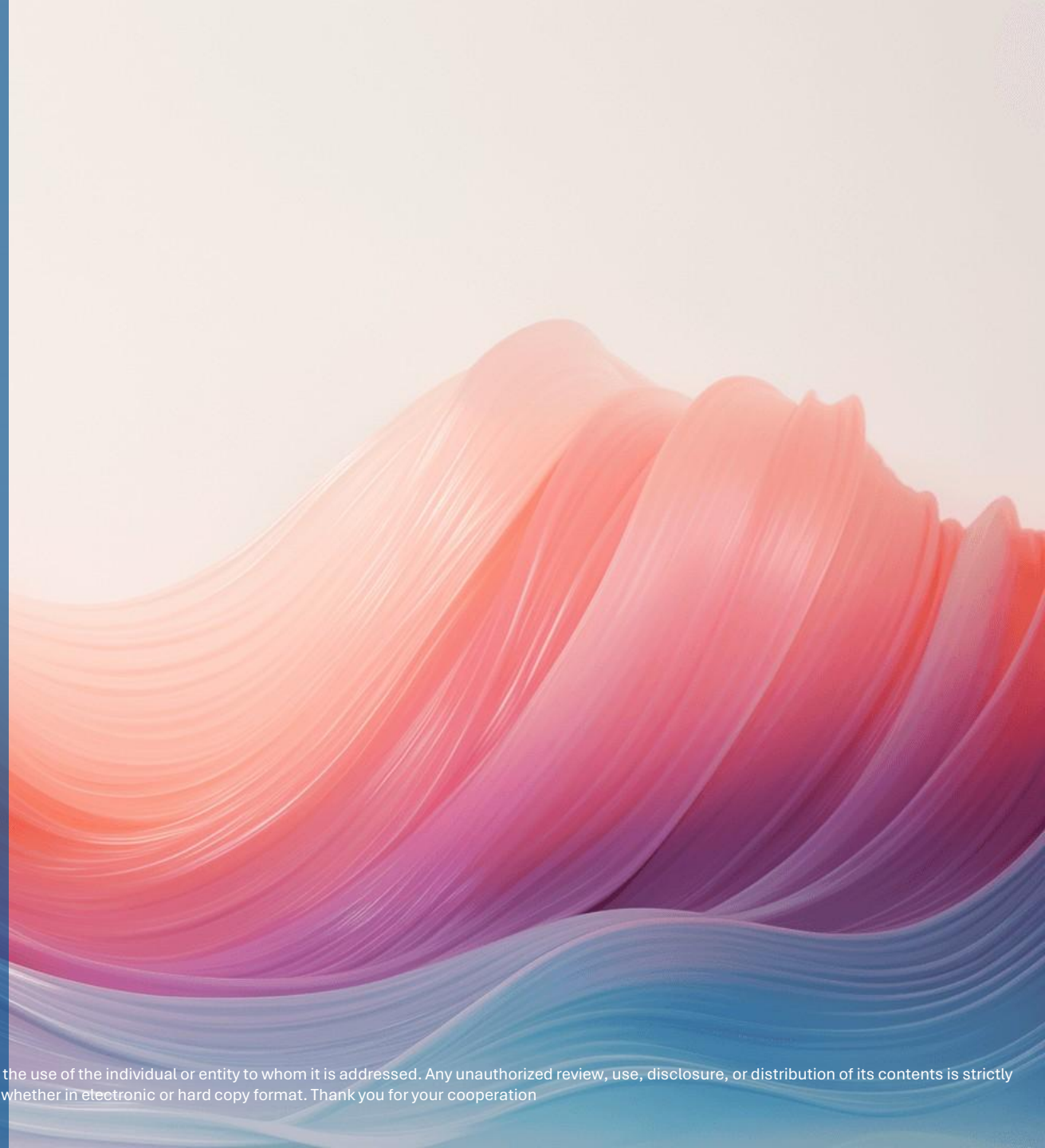
04 Created a reusable, enterprise-scale blueprint designed to extend to 200+ critical APIs across portfolios.

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THANK YOU!

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